

SESSION WILL BE DIVIDED INTO

1. Introduction to Performance Issues in Power BI
2. What is the Performance Analyzer?
3. Open and Use Performance Analyzer
4. Understanding the Outputs
5. **Common optimization techniques**

SECTION 1

Performance Issues In Power BI

Introduction to performance issues

- Why performance matters:
 - User experience: Slow reports = frustrated users.
 - Large datasets and complex models can slow down visuals.
- Common reasons for slowness:
 - Heavy DAX calculations.
 - Inefficient data models.
 - Too many visuals on a page.
 - Complex queries hitting the backend.

SECTION 2

What Is Performance Analyzer Tool

Performance Analyzer Tool

- A built-in tool in Power BI Desktop.
- Purpose:
 - Helps capture **how long each visual** takes to load.
 - Breaks down timings into categories.
- **Three main timings captured:**
 - **DAX Query:** Time spent on querying and running DAX formulas.
 - **Visual Display:** Time taken to render/paint the visual.
 - **Other:** Miscellaneous operations (e.g., background processes).

SECTION 3

How To Access Tool?

Steps to access the tool

1. Go to **View** tab → Click **Performance Analyzer**.
2. The pane opens on the right side.
3. Click **Start Recording**.
4. Refresh the visuals (either page refresh or specific visuals).
5. See timings recorded for each visual.
6. Expand each visual to see the detailed breakdown

SECTION 4

Understanding The output

Output

- **Visual Name:** The name of the visual.
- **Breakdown:**
 - DAX Query Time → Optimization needed if too high.
 - Visual Display Time → May be affected by chart complexity.
 - Other Time → Look here if there's a delay but DAX and Visual are fine.
- **Tip:** Sorting the visuals by highest time helps identify the bottleneck immediately.

SECTION 5

Common Findings And Solution

Issue Found	Solution
High DAX Query Time	Optimize DAX (avoid iterator functions like SUMX if possible, pre-aggregate at model level, use variables wisely).
High Visual Display Time	Simplify visuals (reduce complex charts, avoid excessive use of custom visuals, limit number of visuals per page).
High Other Time	Reduce heavy background processes like tooltips, interactions, sync slicers.